Integer Literals

Examples: 175 123E2 -2 (for 8-bit representation, equivalent to 2#11111110#)
1000_0000_0000 (Underscores are used for increased readability, no change of value)
2#10101111# 8-bit binary representation of 175_{10}
8#257# Octal representation of 175_{10}
16#AF# Hexadecimal representation of 175_{10}

Bit String Literals

Examples: "10101100" 8-bit vector, equivalent to "10_101_100" or "1010_1100" or B"10101100"
O"254" Octal representation of "10101100"
X"AC" Hexadecimal representation of "10101100"

VHDL 2008 enhancements

1. Bit String Literals may contain elements other than '0' or '1'.

   Examples: "1010ZZZZ1000" equivalent to X"AZ8"
   O"3X7" equivalent to "011XXX111" or B"011XXX111"
   X"L5" equivalent to "LLLL0101" or B"LLLL0101"
   D"23" equivalent to "10111" Not available in VHDL 1993

2. Allows Octal bit string length to be NOT of multiple of 3 and Hexadecimal bit string length to be NOT of multiple of 4.

   Examples: 7O"23" equivalent to "0010011"
   6O"23" equivalent to "010011" MSB = '0 truncated
   5O"23" can't truncate non-zero resulted in an error
   6X"4" equivalent to "000100"
   9B"X" equivalent to "00000000X"
   6D"23" equivalent to "010111"

3. Include signed and unsigned number representations : SB, SO, SX, UB, UO, UX

   Examples: 7SX"A" equivalent to "1111010" sign-extended
   7UX"A" equivalent to "0001010" padded with '0'
   8SO"W3" equivalent to "WWWWW011"
   6UB"110" equivalent to "000110" padded with '0'
   6SB"110" equivalent to "111110" sign-extended